

DATA ANALYSIS PROJECT — AMAZON ORDERS DATASET

(Students must use Python + Excel + SQL + Power BI)

Project Overview

You are provided with an Amazon orders dataset (~128000 rows) containing information such as order id, order date, status, categories, size, prices, amount, ship city etc.

Your goal is to clean, analyze, visualize, and interpret the data using **Python**, **Excel**, **SQL**, and **Power BI**.

You must deliver meaningful insights that can help sellers on Amazon, and new entrepreneurs make data-driven decisions.

TOOLS REQUIRED

- **Python** → Data Cleaning, Preprocessing, EDA
 - **Excel** → EDA, Pivot Tables, Initial Exploration
 - **SQL** → Aggregations, Grouped Analysis, Filtering, Multi-table queries
 - **Power BI** → Dashboard building, data modelling, KPIs, interactive visuals
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PROJECT TASKS (TO BE COMPLETED BY STUDENTS)

1. DATA CLEANING & PREPARATION

Perform thorough cleaning of the dataset in Python. Your cleaning steps must be recorded in a separate python file (.py file) named **Data_Cleaning**

Tasks:

1. Identify and handle missing values.
 2. Identify and remove duplicates.
 3. Correct inconsistent formatting in columns (text, numbers, currency, etc.).
 4. Standardize values in categorical columns (status, category, ship-city, etc.).
 5. Treat or flag outliers where necessary.
 6. Create additional derived columns if helpful
 7. Prepare the final cleaned dataset for analysis.
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2. EXPLORATORY DATA ANALYSIS (EDA)

Using Python and Excel, explore the dataset to understand its structure and core characteristics.

Your EDA should address the following:

General Analysis:

- Distribution of Product Category

- Distribution of Product Size
 - Distribution of status
 - Count of orders based on ship-city
 - Count of orders based on ship-state
 - Count of product size based on product category
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3. ANALYTICAL QUESTIONS (STUDENTS MUST ANSWER ALL)

Use python for distribution analysis, SQL for aggregations, Excel for pivots, and Power BI for visualization.

Orders-Based Analysis:

1. Which SKUs, categories, and sizes generate the most net sales?

Metrics: Sales Amount, Quantity

2. Which are the Top 5 states based on Total Order Amount?
 3. Which are the Top 5 cities based on Total Order Amount?
 4. No. of orders based on product category?
 5. 5. No. of orders based on ship-city/ship-state?
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Revenue-Based Analysis:

6. How does revenue trend over time (daily/weekly/monthly)?
7. Which Fulfilment channels (Merchant or Amazon) are most profitable?
8. What is relation between ship-service and Order Amount

9. What is relation between quantity and Amount based on product category
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General Analysis:

10. What is the relation between ship-service and Product Category
 11. What are the cancellation rates by category and size?
 12. What is the relation between type of promotion and order amount?
 13. Category Distribution
 14. Product size distribution
 15. Style Distribution
 16. What percentage of sales comes from top 10 products/categories?
 17. How concentrated is revenue — do a few products dominate, or is it spread across many?
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4. POWER BI DASHBOARD REQUIREMENTS

Create a multi-page dashboard containing:

Page 1 — Overview KPIs

- Total Orders
- Average Order Amount
- % orders based on status
- % orders based on ship-city
- % orders based on category

Page 2 — Orders-Based insights:

- Top 5 states based on orders
- Top 5 cities based on orders
- No. of orders based on courier status
- Daily/Monthly Orders count

Page 3 — Revenue (Order Amount) Insights

- Revenue Trends over time
- Revenue based on ship-service
- Revenue based on ship-city
- Revenue based on ship-state

Page 4 — General Insights

- relation between type of promotion and order amount?
- cancellation rates by category and size
- relation between ship-service and Product Category

Page 5 — Detailed Orders Explorer

- A filterable table of all orders with columns:
status, fulfillment, ship-service, product category, ship-city etc.
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5. PERFORMANCE METRICS (KPIs)

You must calculate and display:

1. Average Order Amount
 2. % orders based on status
 3. % orders based on ship-city
 4. % orders based on category
 5. Top 5 states based on orders
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6. INSIGHTS & STRATEGY REPORT

Prepare a **2–3 page report** explaining:

- Data-driven observations from your analysis
 - Trends in customer orders
 - Popular categories/sizes
 - Factors that influence higher order amounts
 - Suggested business strategies
 - Recommendations for new sellers on Amazon
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7. FINAL SUBMISSION CHECKLIST

Students must submit:

- ✓ Data Cleaning code (python file .py)

- ✓ Cleaned dataset
 - ✓ Excel workbook with pivots & charts
 - ✓ SQL file with all queries
 - ✓ Power BI dashboard (.pbix)
 - ✓ Insights report (PDF)
 - ✓ PowerPoint presentation (optional but recommended)
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